THE COMPARISON MAP BETWEEN PULLBACK AND THE RIGHT ADJOINT IS TRANSITIVE FOR COMPOSABLE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6R (sources/stacks-duality/pinned-original-source.tex line 2189).

Allowed standard prerequisites (editorial).

- The comparison map between the pullback and the right adjoint of push-forward. Same chapter, equation labelled equation-compare-with-pullback, invoked in the statement, and Lemma labelled lemma-twisted-inverse-image for the adjoints a, b, c. Supplies the arrows whose composition is shown to commute.
- Base change for the cohomology of a tensor product with a pullback. Derived Categories of Schemes, Lemma labelled perfect-lemma-cohomology-base-change, cited at proof lines 2217-2220 and 2230-2234. Identifies the source of the adjoint maps and supplies the isomorphism epsilon.
- General properties of adjoint functors, adjunction maps and counits. Categories, Section labelled categories-section-adjoint, cited at proof lines 2250-2252. Gives the identity gamma o f_*delta = alpha o epsilon^{-1} that concludes the proof.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0A9S: sources/stacks-duality/pinned-original-source.tex, source lines 2079–2079. Same-chapter pointer into the author environment equation at source line 2079; invoked at proof line None of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

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Source reference. Original lemma 08EU: sources/stacks-chow/tags.txt, source lines 10685–10685. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2221 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0036: sources/stacks-chow/tags.txt, source lines 123–123. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2250 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ and $g : Y \to Z$ be composable morphisms of quasi-compact and quasi-separated schemes and set $h = g \circ f$. Let a, b, c be the adjoints of Lemma 0A9E for f, g, h. For any $K \in D_{QCoh}(\mathcal{O}_Z)$ the diagram*

$$\begin{array}{ccccc} Lf^*(Lg^*K \otimes^{\mathbf{L}}_{\mathcal{O}_Y} b(\mathcal{O}_Z)) \otimes^{\mathbf{L}}_{\mathcal{O}_X} a(\mathcal{O}_Y) & \longrightarrow & a(Lg^*K \otimes^{\mathbf{L}}_{\mathcal{O}_Y} b(\mathcal{O}_Z)) & \longrightarrow & a(b(K)) \\ \| & & & & \| \\ Lh^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*b(\mathcal{O}_Z) \otimes^{\mathbf{L}}_{\mathcal{O}_X} a(\mathcal{O}_Y) & \longrightarrow & Lh^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} c(\mathcal{O}_Z) & \longrightarrow & c(K) \end{array}$$

is commutative where the arrows are (0A9S) and we have used $Lh^* = Lf^* \circ Lg^*$ and $c = a \circ b$.

Proof. In this proof we will write f_* for Rf_* and f^* for Lf^*, etc, and we will write $\otimes$ for $\otimes^{\mathbf{L}}_{\mathcal{O}_X}$, etc. The composition of the top arrows is adjoint to a map

$$g_*f_*(f^*(g^*K \otimes b(\mathcal{O}_Z)) \otimes a(\mathcal{O}_Y)) \to K$$

The left hand side is equal to $K \otimes g_*f_*(f^*b(\mathcal{O}_Z) \otimes a(\mathcal{O}_Y))$ by Derived Categories of Schemes, Lemma 08EU and inspection of the definitions shows the map comes from the map

$$g_*f_*(f^*b(\mathcal{O}_Z) \otimes a(\mathcal{O}_Y)) \xleftarrow{g_*\epsilon} g_*(b(\mathcal{O}_Z) \otimes f_*a(\mathcal{O}_Y)) \xrightarrow{g_*\alpha} g_*(b(\mathcal{O}_Z)) \xrightarrow{\beta} \mathcal{O}_Z$$

tensored with id_K. Here ϵ is the isomorphism from Derived Categories of Schemes, Lemma 08EU and β comes from the counit map $g_*b \to \mathrm{id}$. Similarly, the composition of the lower horizontal arrows is adjoint to id_K tensored with the composition

$$g_*f_*(f^*b(\mathcal{O}_Z) \otimes a(\mathcal{O}_Y)) \xrightarrow{g_*f_*\delta} g_*f_*(ab(\mathcal{O}_Z)) \xrightarrow{g_*\gamma} g_*(b(\mathcal{O}_Z)) \xrightarrow{\beta} \mathcal{O}_Z$$

where γ comes from the counit map $f_*a \to \mathrm{id}$ and δ is the map whose adjoint is the composition

$$f_*(f^*b(\mathcal{O}_Z) \otimes a(\mathcal{O}_Y)) \xleftarrow{\epsilon} b(\mathcal{O}_Z) \otimes f_*a(\mathcal{O}_Y) \xrightarrow{\alpha} b(\mathcal{O}_Z)$$

By general properties of adjoint functors, adjoint maps, and counits (see Categories, Section 0036) we have $\gamma \circ f_*\delta = \alpha \circ \epsilon^{-1}$ as desired. □